JAMA Psychiatry | Original Investigation

War Exposure and DNA Methylation in Syrian Refugee Children and Adolescents

Demelza Smeeth, PhD; Simone Ecker, PhD; Olga Chervova, PhD; Fiona McEwen, PhD; Elie Karam, MD; Stephan Beck, PhD; Michael Pluess, PhD

IMPORTANCE Exposure to war is associated with poor mental health outcomes. Adverse and traumatic experiences can lead to long-lasting DNA methylation changes, potentially mediating the link between adversity and mental health. To date, limited studies have investigated the impact of war on DNA methylation in children or adolescents, hampering our understanding of the biological impact of war exposure.

OBJECTIVE To identify salivary DNA methylation differences associated with war exposure in refugee children and adolescents.

DESIGN, SETTING, AND PARTICIPANTS This cohort study included Syrian refugee children and adolescents, and their primary caregiver were recruited from tented settlements in Lebanon. Data collection was carried out in 2 waves, 1 year apart, from October 2017 to January 2018 and October 2018 to January 2019. Children and their caregiver were interviewed, and children provided saliva samples for DNA extraction. Data analysis was conducted in 2022, 2023, and 2024.

EXPOSURE War exposure assessed by interviewing children and their caregiver using the War Events Questionnaire.

MAIN OUTCOMES AND MEASURES Salivary DNA methylation levels were assayed with the Infinium MethylationEPIC BeadChip (Illumina). Epigenetic aging acceleration was estimated using a set of preexisting epigenetic aging clocks. A literature search was conducted to identify previously reported DNA methylation correlates of childhood trauma.

RESULTS The study population included 1507 children and adolescents (mean [SD] age, 11.3 [2.4] years; age range, 6-19 years; 793 female [52.6%]). A total of 1449 children provided saliva samples for DNA extraction in year 1, and 872 children provided samples in year 2. Children who reported war events had a number of differentially methylated sites and regions. Enrichment analyses indicated an enrichment of gene sets associated with transmembrane transport, neurotransmission, and intracellular movement in genes that exhibited differential methylation. Sex-stratified analyses found a number of sex-specific DNA methylation differences associated with war exposure. Only 2 of 258 (0.8%) previously reported trauma-associated DNA methylation sites were associated with war exposure (B = −0.004; 95% CI, −0.005 to −0.003; Bonferroni P = .04 and B = −0.005; 95% CI, −0.006 to −0.004; Bonferroni P = .03). Any war exposure or bombardment was nominally associated with decreased epigenetic age using the Horvath multitissue clock (B = −0.39; 95% CI, −0.63 to −0.14; P = .007 and B = −0.42; 95% CI, −0.73 to −0.11; P = .002).

CONCLUSIONS AND RELEVANCE In this cohort of Syrian refugee children and adolescents, war exposure was associated with a small number of distinct differences in salivary DNA methylation.

JAMA Psychiatry. 2025;82(2):191-200. doi:10.1001/jamapsychiatry.2024.3714
Published online November 20, 2024.

Supplemental content

Author Affiliations: Department of Biological and Experimental Psychology, School of Biological and Behavioural Sciences, Queen Mary University of London, London, United Kingdom (Smeeth, McEwen, Pluess); School of Psychology, University of Surrey, Guildford, United Kingdom (Smeeth, Pluess); UCL Cancer Institute, University College London, London, United Kingdom (Ecker, Chervova, Beck); Department of War Studies, King's College London, London, United Kingdom (McEwen); Department of Psychiatry and Clinical Psychology, Balamand University, St Georges Hospital University Medical Center, Institute for Development, Research, Advocacy and Applied Care, Beirut, Lebanon (Karam).

Corresponding Author: Michael Pluess, PhD, School of Psychology, University of Surrey, 05AC05 Lewis Carrol Building, Guildford GU2 7XH, United Kingdom (m.pluess@surrey.ac.uk).

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As of 2023, it was estimated that 117.2 million people had been forcibly displaced worldwide.[1] This population includes a substantial number who had fled countries due to armed conflict.[2] Notably, a large number of these displaced individuals are children,[3] who are at an elevated risk of poor mental health often linked to the experience of premigration war.[4,5] Symptoms of psychological distress are common, and there is a need to better understand the link between war exposure and subsequent mental health.[6,7]

Numerous human studies[8-11] have found that adverse life experiences, ranging from inadequate maternal care to war exposure, can modify biological processes including DNA methylation. DNA methylation is an epigenetic process whereby cytosine-guanine (CpG) dinucleotide sites are chemically modified throughout the genome.[11,12] DNA methylation can control the expression of nearby genes and is a potential mediator between adversity and psychiatric outcomes.[13,14] Early investigations primarily adopted a candidate gene approach, targeting genes involved in key psychiatric pathways, and identified DNA methylation correlates of trauma and adversity.[15-17] More recent, hypothesis-free epigenome-wide association studies (EWAS) have identified DNA methylation differences, although these rarely overlap with candidate loci.[18-20] In addition to studying independent DNA methylation loci, biological or epigenetic aging can be inferred from DNA methylation levels at distinct sets of CpGs.[21-23] Epigenetic aging is frequently exacerbated after adversity.[10,24,25]

To date, there has been limited research concerning the impact of war on DNA methylation, and the majority concentrates on military personnel.[26-28] Civilian populations are rarely studied, and findings are inconclusive; however, war exposure has been consistently associated with global decreases in DNA methylation.[29-31] Most research has been conducted with a small cohort of mother-infant dyads from the Democratic Republic of the Congo. War exposure was associated with DNA methylation in genes associated with neuronal plasticity (*BDNF*) and stress responses (*CRH*, *CRHBP*, *NR3C1*, and *FKBP5*) in various tissues.[32,33] Prenatal war exposure was also associated with DNA methylation in the *NR3C1* promoter in placental tissue but not in maternal blood and vice versa for *IGF1* and *IGF2*.[29] Conversely, a study[34] in older adults found no association between childhood war exposure and *BDNF* DNA methylation. A more recent EWAS[35] on the Congolese dyads found a small number of DNA methylation differences associated with war exposure in both mother and infant as well as accelerated epigenetic aging in infants.

Although there is evidence that many forms of adversity during childhood and adolescence can impact DNA methylation, to our knowledge, only a single study[36] has investigated the impact of war exposure in this age group. This study in Burundian refugee children was unable to detect any buccal DNA methylation differences after war exposure; however, the authors note that it was not adequately powered to detect small effects. Building on this research, we aimed to comprehensively assess the DNA methylation differences associated with war exposure in a sample of Syrian refugee children and adolescents. We aimed to identify differentially methylated probes (DMPs) and differentially methylated regions (DMRs) associated with war exposure in a hypothesis-free manner. In addition, we attempted to replicate previously published DMPs associated with traumatic or adverse experiences in childhood, hypothesizing that these would also be associated with war exposure. Finally, we tested whether epigenetic age acceleration was associated with war exposure, hypothesizing that war would accelerate epigenetic aging.

Key Points

Question What are the salivary DNA methylation differences associated with war exposure in Syrian refugee children and adolescents?

Findings In this cohort study including 1507 Syrian refugee children and adolescents, war exposure was associated with differential methylation at various sites and regions but not at sites previously linked to childhood trauma. Additionally, war exposure was linked to decreased epigenetic aging.

Meaning Results suggest that exposure to war events in refugee children and adolescents was associated with a number of distinct DNA methylation sites.

Methods

Ethical approval for this study was granted by the institutional review board of the University of Balamand/Saint George Hospital University Medical Center, Lebanon. The study was reviewed by the Lebanese National Consultative Committee on Ethics and approved by the Ministry of Public Health. Caregivers provided written informed consent, and children provided assent. This set of analyses has been preregistered.[37] Deviations or additions to the preregistered analysis plan are described. Full methods are detailed in the eMethods in Supplement 1. This study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guidelines.

Sample

This study used a subset of a longitudinal cohort study of 1600 Syrian refugee children and their primary caregivers from informal tented settlements in the Beqaa region of Lebanon.[38] Interview data and saliva samples were collected by trained Arabic-speaking interviewers across 2 study waves, 1 year apart (from October 2017 to January 2018 and October 2018 to January 2019), with pairs followed up. Information on race or ethnicity was not collected based on feedback from local collaborators that the study sample would more likely identify in terms of their religious sect than ethnicity. Instead, we collected information on nationality and religion.

Measures

DNA Methylation

Saliva was collected with Genefix saliva collection tubes (Isohelix). DNA methylation profiling at 865 859 sites was

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conducted on salivary DNA with the Infinium MethylationEPIC v1.0 BeadChip kit (Illumina). Samples from different waves were analyzed concurrently and underwent a series of quality control and normalization steps (eMethods in Supplement 1).

Epigenetic aging was assessed using a range of DNA methylation clocks that were selected for their prevalence in the literature (ie, Horvath multitissue,[22] skin and blood,[39] Hannum,[40] GrimAge[21] and PhenoAge[41]) or their relevance to pediatric populations (PedBE[42] [ie, Pediatric-Buccal-Epigenetic] and Wu clocks[43]) (eTable 1 in Supplement 1) due to the absence of clocks developed specifically for salivary DNA in children. DunedinPACE[44] (ie, Dunedin Pace of Aging Calculated From the Epigenome) was also included due to the use of longitudinal sampling during development.

Interview Data

War exposure was measured using the War Events Questionnaire, a 25-item checklist of war events.[45] We assessed the cumulative effect of war exposure, the impact of different types of exposure, and contrary to the preregistered plan, the impact of any war exposure (eTable 2 in Supplement 1). Age, sex, nationality, smoking status, time since leaving Syria, pubertal status, body mass index (BMI), and the quality of the current living environment were recorded. Genome-wide genotyping data was also derived from salivary DNA.

Statistical Analysis

Analyses were conducted using R, version 4.2.2, in RStudio. DMPs associated with war exposure were identified by constructing probe-wise linear models with Limma,[46] restricted to variable autosomal probes with greater than 5% B value ranging between the 10th and 90th percentile across all individuals (544 587 probes). Bonferroni-corrected $\alpha = .05$ was used as a significance threshold, and all P values were 2-sided. Gene set enrichment analyses were performed using the methylglm function from the methylGSA package,[47] which accounts for the unequal number of probes per gene without the need to specify a P value cutoff. Enriched gene sets were considered where false discovery rate P value was <.05. DMRs were identified with DMRcate, where Stouffer P value was <.05.[48]

Replication analyses were conducted on CpGs that had been previously associated with traumatic experiences in childhood. Probe-wise linear mixed-effects models were fitted using the lme4 package[49] using the repeated samples for each individual where available. We considered successful replication where the difference in DNA methylation was in the same direction as the original report and Bonferroni P <.05.

Epigenetic age analyses were only conducted on those with the highest confidence in the accuracy of chronological age. For each clock, epigenetic age acceleration was calculated as the residual of a linear mixed-effects model that fitted epigenetic age onto chronological age, alongside covariates. Analyses tested the corrected residuals within linear mixed-effects models with war exposure and sex as fixed effects and the donor as a random effect. Data analysis was conducted in 2022, 2023, and 2024.

Table. Cohort Description at Baseline

Characteristic	No. (%)
Total No.	1507
Sex, No. (%)	
Female	793 (52.6)
Male	714 (47.4)
Age, mean (SD), y	11.3 (2.4)
Nationality, No. (%)	
Syrian	1485 (98.5)
Other[a]	22 (1.5)
Family religion, No. (%)[b]	
Catholic	1 (0.1)
Shia	6 (0.4)
Sunni	1460 (96.9)
None	13 (0.9)
Other	2 (0.1)
Reported smoker, No. (%)	19 (1.3)
BMI, mean (SD)[c]	17.9 (3.6)
Time since leaving Syria, No. (%), mo[d]	
0-12	275 (18.2)
12-24	213 (14.1)
24-36	212 (14.1)
36-48	562 (37.3)
>48	239 (15.9)
Any war exposure, No. (%)	1462 (97.0)
Bombardment, No. (%)	1434 (95.2)
Other-directed violence, No. (%)	1160 (77.0)
Violence directed toward a close person, No. (%)	853 (56.6)
War-related violence in the home, No. (%)	995 (66.0)
Personal harm, No. (%)	453 (30.1)
Total war exposure, mean (SD)	9.6 (5.5)

Abbreviation: BMI, body mass index.

[a] Other nationality includes Iraqi, Lebanese, and Palestinian.

[b] A total of 25 participants were missing family religion; other religion is a religion other than Alawite, Catholic, Druze, Maronite, Orthodox, Shia, or Sunni.

[c] There were 643 missing BMI, which is calculated as weight in kilograms divided by height in meters squared.

[d] There were 6 missing time since leaving Syria.

Results

Epigenome-Wide DNA Methylation Differences in War-Exposed Children

After exclusions, the study population included a total of 1507 children and adolescents (1449 youth in year 1 and 872 youth in year 2; mean [SD] age, 11.3 [2.4] years; age range, 6-19 years; 793 female [52.6%]; 714 male [47.4%]) (**Table**). Families were primarily Syrian, although 1.5% (22 of 1507) reported other nationalities, which included Iraqi, Lebanese, and Palestinian. A total of 96.9% of the sample (1460 of 1507) were Sunni Muslim, the religious majority in Syria.

We first aimed to identify probes and regions of the genome that were differentially methylated with exposure to war

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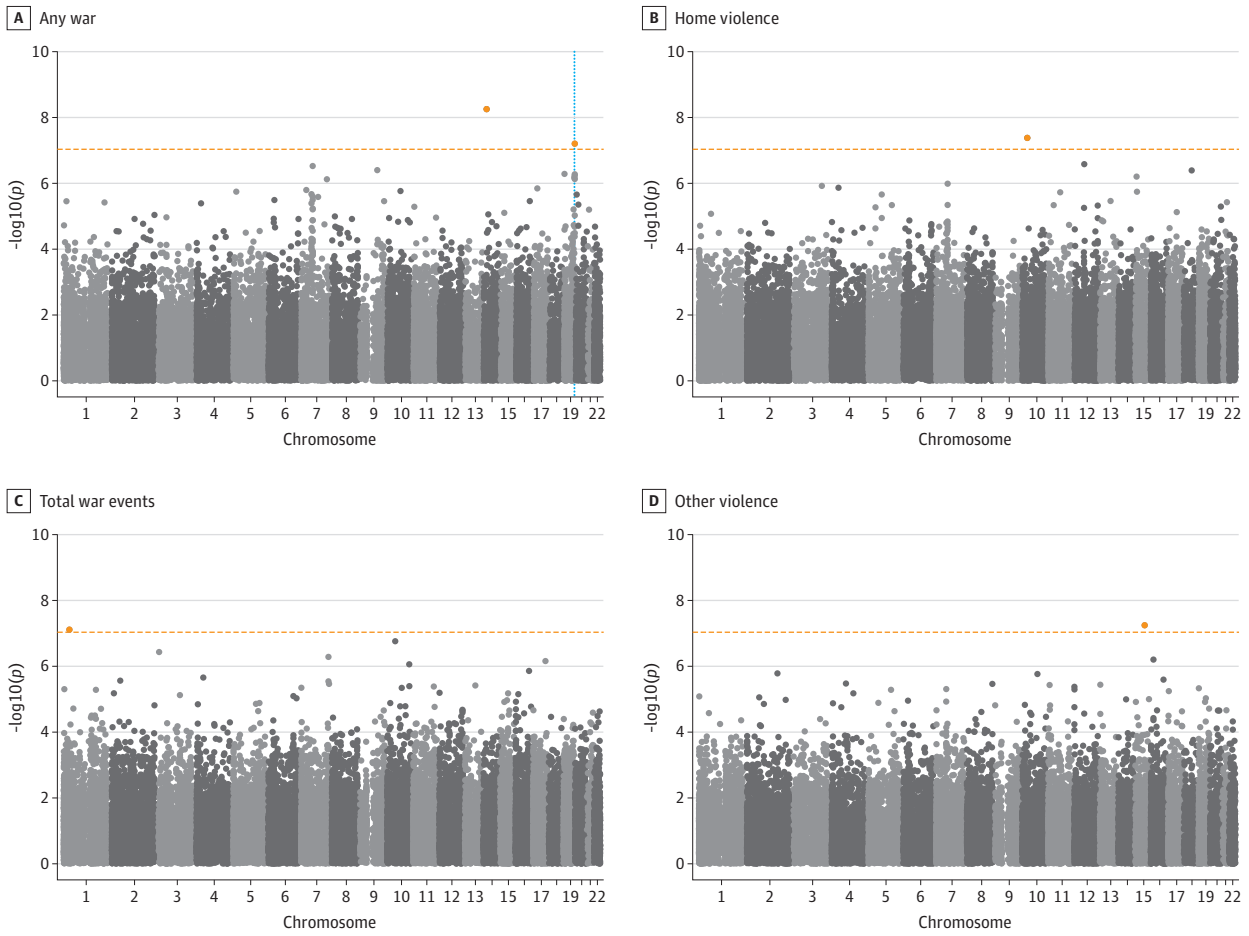
Figure 1. Manhattan Plots of the Epigenome-Wide Association Studies for Violence in the Home, Any War Exposure, Total War Exposure, and Other Forms of Violence



The orange dashed line indicates the Bonferroni-corrected *P* value threshold. Genome-wide significant differentially methylated probes are in orange and the only differentially methylated region is indicated by the blue vertical dotted line.

in a hypothesis-free manner. We found that children who reported war events that impacted their home exhibited increased DNA methylation at cg18691565 (Bonferroni *P* = .02; *ACBD5*), those who reported any war exposure exhibited reduced methylation at cg08095654 (Bonferroni *P* = .03) and cg03806695 (Bonferroni *P* = .003), and those who reported other forms of war-related violence exhibited reduced methylation at cg14747961 (Bonferroni *P* = .03; *SCAMP5*) (**Figure 1** and eTable 3 and eFigure 2 in Supplement 1). The number of war-related events experiences was positively associated with methylation at cg17049090 (Bonferroni *P* = .04; *ADGRB2*). No inflation in the test statistic was observed when considering any war exposure or violence in the home (λ = 0.99 and λ = 0.95, respectively) (eFigure 3 in Supplement 1). However, there was some evidence for inflation when considering other war-related violence or total war events, suggesting caution when interpreting this result (λ = 1.19 and λ = 1.14, respectively). Follow-up analyses indicated that these DNA methylation differences were not confounded by analysis batch, time since leaving Syria, pubertal status, BMI, the quality of the current living environment, or population stratification nor were they biased by genetic relatedness across the sample (eTable 4 in Supplement 1).

Exploratory enrichment analyses of the results were performed to identify pathways or biological functions that are overrepresented in the top DMPs. These indicated an enrichment of DNA methylation differences in gene sets related to transmembrane transport (amino acid: sodium symporter activity) associated with total war exposure, protein degradation (proteasome accessory complex) in those who experience bombardment, and neurotransmission (glutamate receptor activity) and intracellular movement (microfilament motor activity) in those who experienced violence to a close individual (**Figure 2** and eTable 5 in Supplement 1). We only identified a significant DMR in those who reported any war exposure (chromosome 19: 58728390-58728865) (Figure 1A and eTable 6 in Supplement 1). This DMR contained 4 of the top 15 DMPs associated with any war exposure; however, it has not been annotated to a gene.

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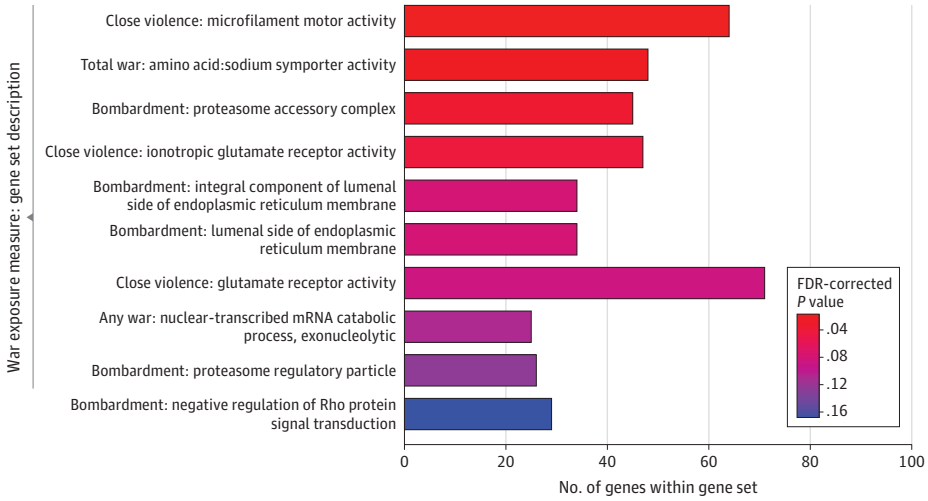
Figure 2. Top 10 Gene Sets Associated With War Exposure



Each gene set is labeled with the description and war exposure measure with which it is associated. *P* values have been false discovery rate (FDR)–corrected. Bar length represents the number of genes within that gene set. mRNA indicates messenger RNA.

Sex-stratified analyses were performed to identify sex-specific DMPs and DMRs, as well as those located on the sex chromosomes. In females, cg09582238 (*PERM1*) was associated with bombardment, cg18518909 (*ATP6V1H*) was associated with other forms of violence, cg20560283 was associated with violence in the home, and 6 CpGs (cg12801791, *TESC*; cg00588499; cg18251449; cg25844655; cg11571585; cg11571585, *GSX1*) were associated with any war exposure (eTable 7 in Supplement 1). In males, cg11337624 was correlated with total war exposure, cg13647973 was associated with any war exposure, and 3 CpGs (cg13647973; cg22321036; cg04673462) were associated with bombardment (eTable 8 in Supplement 1). The majority of these DMPs exhibited significant interactions with sex when considering the entire cohort. In females, exposure to any war-related events was associated with 4 DMRs (chromosome 19: 58728390-58728865; chromosome 17: 21187376-21187420; chromosome 2: 120516537-120516628; chromosome 7: 63386226-63387147, *RP11-73B2.6*) (eTable 9 in Supplement 1), whereas bombardment was associated with a single DMR (chromosome 4: 124232-125504, *ZNF718*). In males, exposure to bombardment was associated with a single DMR (chromosome 1: 38460950-38461896) (eTable 10 in Supplement 1).

Trauma-Associated DNA Methylation Differences in War-Exposed Children

We next aimed to replicate existing trauma-associated DMPs. We identified a total of 267 (258 autosomal) CpGs from the literature that were associated with potentially traumatic experiences in childhood. Of these, only 2 CpGs (0.8%; cg24244000 and cg05717473) were associated with war exposure (violence in the home and other forms of violence) in the current sample after correction for multiple testing (B = −0.004; 95% CI, −0.005 to −0.003; *P* <.001; Bonferroni *P* = .04 and B = −0.005; 95% CI, −0.006 to −0.004; *P* <.001; Bonferroni *P* = .03) (**Figure 3**). Furthermore, most were not significant before Bonferroni corrections (83 of 1803 [4.6%]), and of those that were, only 43% (36 of 83) had an identical direction of effect to the original report (eTable 11 in Supplement 1).

In females, a single CpG (cg10091102) exhibited reduced DNA methylation in those who reported any war exposure (B = −0.008; 95% CI, −0.010 to −0.007; *P* <.001; Bonferroni *P* = .002) or bombardment (B = −0.006; 95% CI, −0.007 to −0.005; *P* <.001; Bonferroni *P* = .005) (eFigure 4A in Supplement 1); however, the direction of effect opposed that of the original report.[50] There were no replicated DMPs in males (eFigure 4B in Supplement 1).

Epigenetic Aging Differences in War-Exposed Children

Finally, we investigated the association of war exposure with age acceleration using commonly used epigenetic aging clocks. Epigenetic age analyses were only conducted on those with the highest confidence in the accuracy of chronological age (year 1: n = 994; year 2: 571). Overall, epigenetic ages were positively correlated with chronological age although correlation strengths varied between clocks (eFigure 1 in Supplement 1).

Horvath's multitissue clock indicated reduced epigenetic aging in those that reported bombardment-related events (B = −0.39; 95% CI, −0.63 to −0.14; *P* = .002) or any war exposure (B = −0.42; 95% CI, −0.73 to −0.11; *P* = .007) (**Figure 4**). However, these do not survive when applying correction for multiple testing using the estimated effective number of tests estimated due to nonindependence (Bonferroni *P* = .08 and Bonferroni *P* = .26, respectively). These results were similar when controlling for a number of potential confounders (eFigure 5 in Supplement 1). When clocks developed in children were applied to the least developmentally advanced of the cohort, age acceleration using Horvath's clock was again negatively associated with bombardment (B = −0.37; 95% CI, −0.70 to −0.63; *P* = .006) and any war exposure (B = −0.37; 95% CI, −0.70 to −0.05; *P* = .02) (eFigure 6 in Supplement 1).

Sex-stratified analyses revealed similar associations between war exposure and aging but differing indicative clocks (eFigure 7 in Supplement 1). In females, exposure to any war-related events (B = −0.52; 95% CI, −0.91 to −0.13; *P* = .01) and

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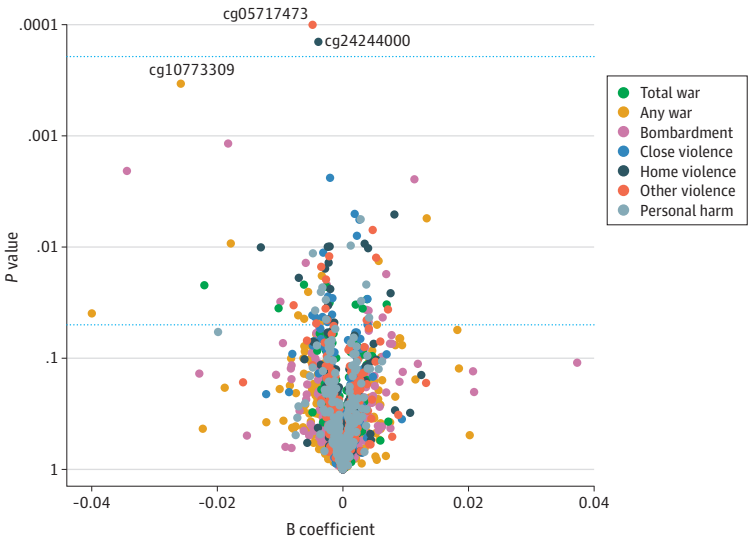
Figure 3. Volcano Plot of the Regression B Values Against *P* Values for the Replication Analyses



Dashed lines represent the nominal (*P* = .05) and Bonferroni-corrected *P* value thresholds. Cytosine-guanine dinucleotide sites with *P* < .001 have been labeled.

bombardment (B = −0.38; 95% CI, −0.70 to −0.07; *P* = .02) was associated with decreased age-acceleration according to Horvath's clock, whereas bombardment was associated with decreased aging according to Wu's clock in males (B = −0.44; 95% CI, −0.83 to −0.04; *P* = .03). Despite these differences, there were no significant interactions with sex.

Discussion

In this cohort study, our epigenome-wide approach revealed a small number of DMPs and DMRs associated with total war exposure, any war exposure, violence in the home, and other forms of war-related violence. To our knowledge, DNA methylation at these loci have not been previously associated with similar exposures, although 1 CpG was annotated to a gene that has been shown to modulate susceptibility to depressivelike behaviors in a mouse model (*ADGRB2*).[51]

We also identified a number of sex-specific DMPs that have been associated with aging (cg18518909, cg20560283, cg12801791, cg18251449) or annotated to genes associated with Gulf War illness (*TESC*, *ATP6V1H*)[52] or neuronal development (*GSX1*).[53] Gene-enrichment analyses suggested an overrepresentation of DNA methylation differences in genes associated with transmembrane transport and neurotransmission among others. However, caution is required given that many were associated with war exposure variables lacking significant DMPs and DMRs. Additionally, the biological significance of any identified DNA methylation differences is unclear (change of 2.4%, 4.3% and 2.0% for the main binary DMPs). Although DNA methylation differences below 5% are considered small, these are in line with existing studies looking at the impact of environmental adversities within peripheral tissues.[54,55] Furthermore, they are only small when considering the cell population as a whole. At a cellular level, the difference between unmethylated and methylated may have significant impacts on that cell's functioning.[56] This is particularly relevant when considering heterogeneous tissues like saliva. Although enrichment analyses provide some clue as to the downstream impacts of such DNA methylation differences, further multiomic or functional investigation would be needed to elucidate the true outcomes.

Due to a lack of similar cohorts for replication, we attempted to replicate previously published CpGs associated with childhood traumatic experiences. There was little evidence that these were associated with war exposure in our cohort with only 2 DMPs surviving multiple testing corrections. There are various reasons why replication failed. First, research has focused predominately on White European populations. Some adversity-associated DNA methylation loci are genotype dependent,[57-59] potentially driving population-level disparities. Second, war exposure may represent a unique form of adversity. Existing research tends to examine a broad set of adversities that include more common and less life-threatening events such as bullying or poverty. Finally, this study measured DNA methylation in saliva, which is a highly heterogeneous tissue and rarely used within the studies identified for replication. Although adversity-associated DNA methylation differences may be mirrored across multiple tissues, they can differ significantly.[18]

Despite the selected epigenetic clocks having been trained predominately in adult populations in tissues other than saliva, many were moderately correlated with chronological age. Interestingly, the 2 clocks (Horvath and Wu), which were nominally associated with war exposure, were some of the few clocks developed using pediatric samples indicating the importance of carefully selecting the most appropriate clock for the study population and tissue. Consequently, these clocks had marginally better agreement with chronological age, going some way to explain these findings. Furthermore, Horvath's multitissue clock was developed on a particularly large sample of 51 healthy tis-

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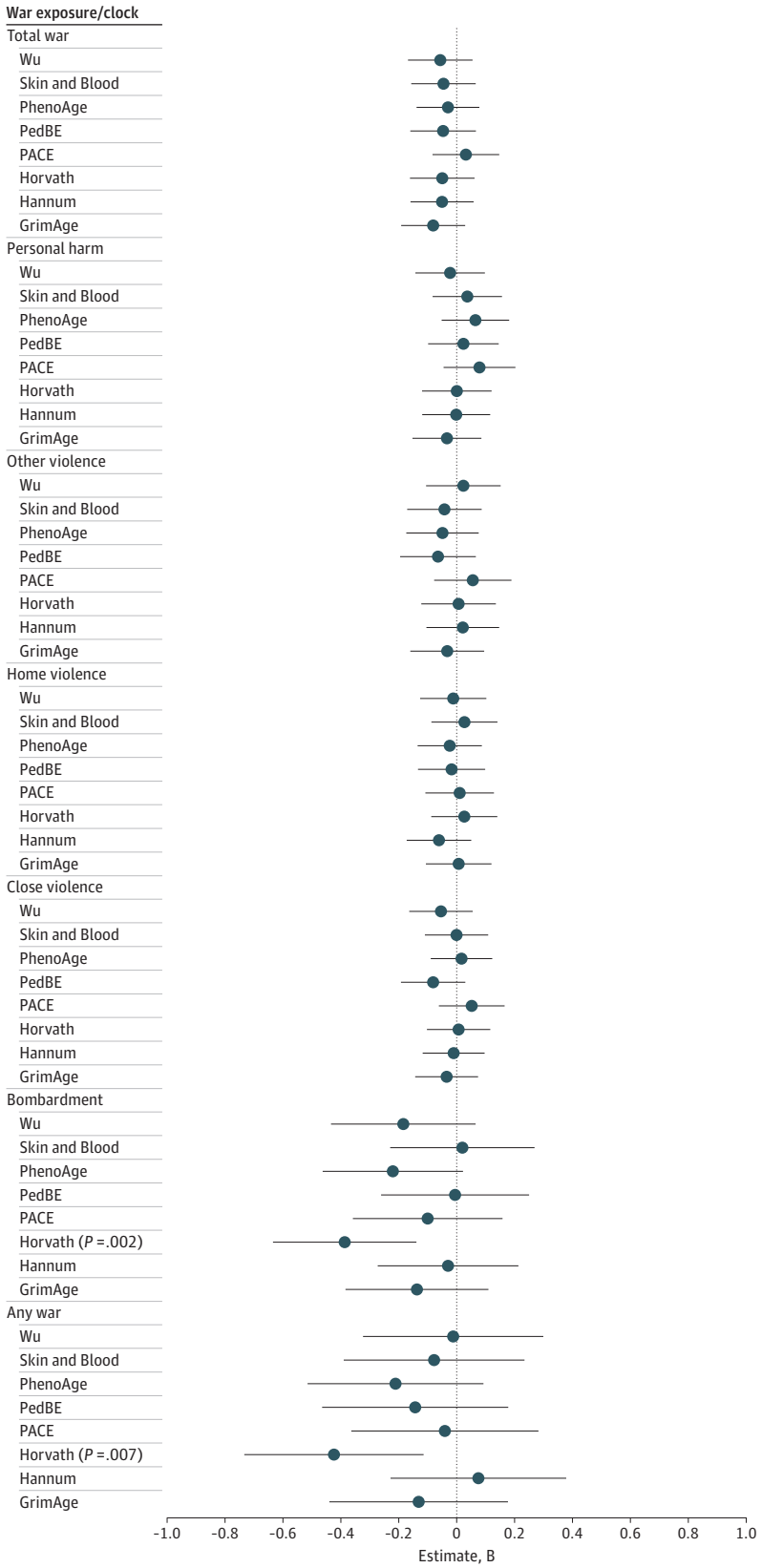
Figure 4. Results for the Association Between Epigenetic Age Acceleration and War Exposure



Dots represent regression coefficients and error bars represent 95% confidence intervals. Negative estimates indicate slower age acceleration and positive estimates indicate faster age acceleration. Where $P < .05$, the P value is given. PACE indicates DunedinPACE or Dunedin Pace of Aging Calculated From the Epigenome; PedBE, Pediatric-Buccal-Epigenetic.

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sues and cell types allowing its application to multiple tissues. The Wu clock was developed on a predominantly male sample,[43] potentially explaining why it was only associated with war exposure in males.

Contrary to our hypotheses and most existing research,[60-63] we found that war exposure was nominally associated with decreased epigenetic aging. However, a recent study[64] suggests that the direction of adversity-associated aging may differ within populations with a relatively high burden of adversity such as care leavers. Similarly, this cohort has been forcibly displaced and are subject to continued adversity.[5,65] These are important to consider when accumulation of stressors and more immediate factors in the lives of refugees may have a greater impact on mental health.[66,67] Alternatively, earlier exposure to war may influence their current environments (eg, increased parental support), tempering epigenetic age acceleration.[68-70]

It should be noted that aging is an unclear concept within younger populations. Although in adults epigenetic aging is considered detrimental and is associated with degeneration and mortality, in children it may simply represent development.[71] Accordingly, our preliminary findings may suggest that war exposure is associated with delayed development in a similar manner to the observed association between traumatic experiences and delayed physical and cognitive development.[72-74] However, it should be emphasized that the observed age deceleration did not survive multiple correction and that some epigenetic clocks performed poorly within this cohort. Future replication or the development of better-performing epigenetic clocks for our samples may help support this finding.

Limitations

This study has some limitations. Despite attempts to capture the severity of war exposure through the use of various war exposure measures, it is likely that this approach does not fully appreciate the complexity of war. It does not capture the repetition or chronicity of each war event, nor does it capture the subjective assessment of the individual. Furthermore, we have no information on the timing of such events, which precludes investigating the stability of any DNA methylation differences identified here. Second, the wide age range of our sample as well as the differences in timing of exposure may hinder the identification of DNA methylation differences, which are age or timing specific. Finally, our sample contains a relatively small number of individuals who reported no war exposure.

Conclusions

To our knowledge, this was only the second study to examine the association of postnatal war exposure with DNA methylation in children or adolescents and one of only a few investigations in a civilian cohort. In addition to furthering our understanding of the biological impact of war, this cohort study also provides much-needed research on underresearched populations outside of Europe and North America. War exposure was associated with various differences in DNA methylation in Syrian refugee children and adolescents, supporting the biological embedding of negative life events. There is little evidence that these overlap with previously reported adversity-associated DNA methylation phenotypes, but findings suggest that war-associated adversity in children and adolescents may delay development.

ARTICLE INFORMATION

Accepted for Publication: September 16, 2024.

Published Online: November 20, 2024. doi:10.1001/jamapsychiatry.2024.3714

Author Contributions: Dr Smeeth and Prof Pluess had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.
Concept and design: Smeeth, Chervova, Karam, Pluess.
Acquisition, analysis, or interpretation of data: All authors.
Drafting of the manuscript: Smeeth, Pluess.
Critical review of the manuscript for important intellectual content: All authors.
Statistical analysis: Smeeth, Ecker, Chervova, Karam.
Obtained funding: Karam, Pluess.
Administrative, technical, or material support: Smeeth, McEwen, Karam.
Supervision: Ecker, Karam, Beck, Pluess.

Conflict of Interest Disclosures: None reported.

Funding/Support: The BIOPATH study was funded by the Eunice Shriver National Institute of Child Health & Human Development (R01HD083387). Training was provided by the University of Michigan supported by a grant from the National Institutes of Health National Institute on Aging (R25AG053227).

Role of the Funder/Sponsor: The funders had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Data Sharing Statement: See Supplement 2.

Additional Contributions: We thank all participating families for their participation; Cassandra M. Popham, PhD (Queen Mary University of London), Andrew May, PhD (Queen Mary University of London), Patricia Moghames, MSc, Stephanie Legoff, MSc, Nicolas Puvis, MSc, and Zeina Hassan, MSc, and all other members of the BIOPATH team for their dedication, hard work, and insights; the Genome Centres at Queen Mary University London and University College London for the DNA extraction and DNA methylation analysis. Beyond usual salary, no one received financial compensation for their contributions.

Additional Information: This article is dedicated to John Fayyad, MD, who sadly died during the study.

REFERENCES

1. United Nations High Commissioner for Refugees. 2023. Global focus. Accessed January 7, 2024. https://reporting.unhcr.org/globalappeal-2023

2. United Nations High Commissioner for Refugees. Figures at a glance. Accessed August 20, 2023. https://www.unhcr.org/uk/about-unhcr/who-we-are/figures-glance

3. UNICEF. Child displacement. Accessed August 20, 2023. https://data.unicef.org/topic/child-migration-and-displacement/displacement/

4. Brandt L, Henssler J, Müller M, Wall S, Gabel D, Heinz A. Risk of psychosis among refugees: a systematic review and meta-analysis. *JAMA Psychiatry*. 2019;76(11):1133-1140. doi:10.1001/jamapsychiatry.2019.1937

5. Guo L, Li L, Xu K, et al. Characterization of premigration and postmigration multidomain factors and psychosocial health among refugee children and adolescents after resettlement in Australia. *JAMA Netw Open*. 2023;6(4):e235841. doi:10.1001/jamanetworkopen.2023.5841

6. Bürgin D, Anagnostopoulos D, Vitiello B, Sukale T, Schmid M, Fegert JM; Board and Policy Division of ESCAP. Impact of war and forced displacement on children's mental health-multilevel, needs-oriented, and trauma-informed approaches. *Eur Child Adolesc Psychiatry*. 2022;31(6):845-853. doi:10.1007/s00787-022-01974-z

7. Frounfelker RL, Miconi D, Farrar J, Brooks MA, Rousseau C, Betancourt TS. Mental health of refugee children and youth: epidemiology, interventions, and future directions. *Annu Rev Public Health*. 2020;41:159-176. doi:10.1146/annurev-publhealth-040119-094230

8. Vinkers CH, Kalafateli AL, Rutten BPF, et al. Traumatic stress and human DNA methylation:

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a critical review. *Epigenomics*. 2015;7(4):593-608. doi:10.2217/epi.15.11

9. Szyf M. The epigenetics of perinatal stress. *Dialogues Clin Neurosci*. 2019;21(4):369-378. doi:10.31887/DCNS.2019.21.4/mszyf

10. Wolf EJ, Maniates H, Nugent N, et al; Va Mid-Atlantic Mirecc Workgroup. Traumatic stress and accelerated DNA methylation age: a meta-analysis. *Psychoneuroendocrinology*. 2018;92:123-134. doi:10.1016/j.psyneuen.2017.12.007

11. Aristizabal MJ, Anreiter I, Halldorsdottir T, et al. Biological embedding of experience: a primer on epigenetics. *Proc Natl Acad Sci U S A*. 2020;117(38):23261-23269. doi:10.1073/pnas.1820838116

12. Allis CD, Jenuwein T. The molecular hallmarks of epigenetic control. *Nat Rev Genet*. 2016;17(8):487-500. doi:10.1038/nrg.2016.59

13. Copeland WE, Shanahan L, Hinesley J, et al. Association of childhood trauma exposure with adult psychiatric disorders and functional outcomes. *JAMA Netw Open*. 2018;1(7):e184493. doi:10.1001/jamanetworkopen.2018.4493

14. Thumfart KM, Jawaid A, Bright K, Flachsmann M, Mansuy IM. Epigenetics of childhood trauma: long-term sequelae and potential for treatment. *Neurosci Biobehav Rev*. 2022;132:1049-1066. doi:10.1016/j.neubiorev.2021.10.042

15. Harms MB, Birn R, Provencal N, et al. Early life stress, FK506 binding protein 5 gene (*FKBP5*) methylation, and inhibition-related prefrontal function: a prospective longitudinal study. *Dev Psychopathol*. 2017;29(5):1895-1903. doi:10.1017/S095457941700147X

16. Peng H, Zhu Y, Strachan E, et al. Childhood trauma, DNA methylation of stress-related genes, and depression: findings from 2 monozygotic twin studies. *Psychosom Med*. 2018;80(7):599-608. doi:10.1097/PSY.0000000000000604

17. Zou Z, Huang Y, Wang J, Min W, Zhou B. DNA methylation of IL-4 gene and the association with childhood trauma in panic disorder. *Psychiatry Res*. 2020;293:113385. doi:10.1016/j.psychres.2020.113385

18. Kandaswamy R, Hannon E, Arseneault L, et al. DNA methylation signatures of adolescent victimization: analysis of a longitudinal monozygotic twin sample. *Epigenetics*. 2021;16(11):1169-1186. doi:10.1080/15592294.2020.1853317

19. Marzi SJ, Sugden K, Arseneault L, et al. Analysis of DNA methylation in young people: limited evidence for an association between victimization stress and epigenetic variation in blood. *Am J Psychiatry*. 2018;175(6):517-529. doi:10.1176/appi.ajp.2017.17060693

20. Houtepen LC, Hardy R, Maddock J, et al. Childhood adversity and DNA methylation in 2 population-based cohorts. *Transl Psychiatry*. 2018;8(1):266. doi:10.1038/s41398-018-0307-3

21. Lu AT, Quach A, Wilson JG, et al. DNA methylation GrimAge strongly predicts lifespan and health span. *Aging (Albany NY)*. 2019;11(2):303-327. doi:10.18632/aging.101684

22. Horvath S. DNA methylation age of human tissues and cell types. *Genome Biol*. 2013;14(10):R115. doi:10.1186/gb-2013-14-10-r115

23. Horvath S, Raj K. DNA methylation-based biomarkers and the epigenetic clock theory of ageing. *Nat Rev Genet*. 2018;19(6):371-384. doi:10.1038/s41576-018-0004-3

24. Boks MP, van Mierlo HC, Rutten BPF, et al. Longitudinal changes of telomere length and epigenetic age related to traumatic stress and post-traumatic stress disorder. *Psychoneuroendocrinology*. 2015;51:506-512. doi:10.1016/j.psyneuen.2014.07.011

25. Clausing ES, Binder AM, Non AL. Epigenetic age associates with psychosocial stress and resilience in children of Latinx immigrants. *Epigenomics*. 2021;13(21):1677-1699. doi:10.2217/epi-2019-0343

26. Hossack MR, Reid MW, Aden JK, Gibbons T, Noe JC, Willis AM. Adverse childhood experience, genes, and PTSD risk in soldiers: a methylation study. *Mil Med*. 2020;185(3-4):377-384. doi:10.1093/milmed/usz292

27. Snijders C, Maihofer AX, Ratanatharathorn A, et al; PGC PTSD EWAS Consortium. Longitudinal epigenome-wide association studies of 3 male military cohorts reveal multiple CpG sites associated with post-traumatic stress disorder. *Clin Epigenetics*. 2020;12(1):11. doi:10.1186/s13148-019-0798-7

28. Straight B, Fisher G, Needham BL, et al. Lifetime stress and war exposure timing may predict methylation changes at NR3C1 based on a pilot study in a warrior cohort in a small-scale society in Kenya. *Am J Hum Biol*. 2021;33(4):e23515. doi:10.1002/ajhb.23515

29. Rodney NC, Mulligan CJ. A biocultural study of the effects of maternal stress on mother and newborn health in the Democratic Republic of Congo. *Am J Phys Anthropol*. 2014;155(2):200-209. doi:10.1002/ajpa.22568

30. Clukay CJ, Hughes DA, Rodney NC, Kertes DA, Mulligan CJ. DNA methylation of methylation complex genes in relation to stress and genome-wide methylation in mother-newborn dyads. *Am J Phys Anthropol*. 2018;165(1):173-182. doi:10.1002/ajpa.23341

31. Montoya-Williams D, Quinlan J, Clukay C, Rodney NC, Kertes DA, Mulligan CJ. Associations between maternal prenatal stress, methylation changes in IGF1 and IGF2, and birth weight. *J Dev Orig Health Dis*. 2018;9(2):215-222. doi:10.1017/S2040174417000800

32. Kertes DA, Bhatt SS, Kamin HS, Hughes DA, Rodney NC, Mulligan CJ. *BNDF* methylation in mothers and newborns is associated with maternal exposure to war trauma. *Clin Epigenetics*. 2017;9:68. doi:10.1186/s13148-017-0367-x

33. Kertes DA, Kamin HS, Hughes DA, Rodney NC, Bhatt S, Mulligan CJ. Prenatal maternal stress predicts methylation of genes regulating the hypothalamic-pituitary-adrenocortical system in mothers and newborns in the Democratic Republic of Congo. *Child Dev*. 2016;87(1):61-72. doi:10.1111/cdev.12487

34. Zhou A, Ancelin ML, Ritchie K, Ryan J. Childhood adverse events and *BDNF* promoter methylation in later-life. *Front Psychiatry*. 2023;14:1108485. doi:10.3389/fpsyt.2023.1108485

35. Quinn EB, Hsiao CJ, Maisha FM, Mulligan CJ. Prenatal maternal stress is associated with site-specific and age acceleration changes in maternal and newborn DNA methylation. *Epigenetics*. 2023;18(1):2222473. doi:10.1080/15592294.2023.2222473

36. Mattonet K, Scharpf F, Block K, Kumsta R, Hecker T. No association between war-related trauma or PTSD symptom severity and epigenome-wide DNA methylation in Burundian refugees. *Eur J Psychotraumatol*. 2023;14(2):2228155. doi:10.1080/20008066.2023.2228155

37. OSF Registries. Associations between salivary DNA methylation and war exposure in Syrian refugee children and adolescents. Accessed August 1, 2023. https://osf.io/6u27r

38. McEwen FS, Popham C, Moghames P, et al. Cohort profile: biological pathways of risk and resilience in Syrian refugee children (BIOPATH). *Soc Psychiatry Psychiatr Epidemiol*. 2022;57(4):873-883. Published online January 18, 2022. doi:10.1007/s00127-022-02228-8

39. Horvath S, Oshima J, Martin GM, et al. Epigenetic clock for skin and blood cells applied to Hutchinson Gilford progeria syndrome and ex vivo studies. *Aging (Albany NY)*. 2018;10(7):1758-1775. doi:10.18632/aging.101508

40. Hannum G, Guinney J, Zhao L, et al. Genome-wide methylation profiles reveal quantitative views of human aging rates. *Mol Cell*. 2013;49(2):359-367. doi:10.1016/j.molcel.2012.10.016

41. Levine ME, Lu AT, Quach A, et al. An epigenetic biomarker of aging for lifespan and health span. *Aging (Albany NY)*. 2018;10(4):573-591. doi:10.18632/aging.101414

42. McEwen LM, O'Donnell KJ, McGill MG, et al. The PedBE clock accurately estimates DNA methylation age in pediatric buccal cells. *Proc Natl Acad Sci U S A*. 2020;117(38):23329-23335. doi:10.1073/pnas.1820843116

43. Wu X, Chen W, Lin F, et al. DNA methylation profile is a quantitative measure of biological aging in children. *Aging (Albany NY)*. 2019;11(22):10031-10051. doi:10.18632/aging.102399

44. Belsky DW, Caspi A, Corcoran DL, et al. DunedinPACE, a DNA methylation biomarker of the pace of aging. *Elife*. 2022;11:e73420. doi:10.7554/eLife.73420

45. Karam EG, al-Atrash R, Saliba S, Melhem N, Howard D. The war events questionnaire. *Soc Psychiatry Psychiatr Epidemiol*. 1999;34(5):265-274. doi:10.1007/s001270050143

46. Ritchie ME, Phipson B, Wu D, et al. Limma powers differential expression analyses for RNA-sequencing and microarray studies. *Nucleic Acids Res*. 2015;43(7):e47. doi:10.1093/nar/gkv007

47. Ren X, Kuan PF. methylGSA: A bioconductor package and shiny app for DNA methylation data length bias adjustment in gene set testing. *Bioinformatics*. 2019;35(11):1958-1959. doi:10.1093/bioinformatics/bty892

48. Peters TJ, Buckley MJ, Statham AL, et al. De novo identification of differentially methylated regions in the human genome. *Epigenetics Chromatin*. 2015;8(1):6. doi:10.1186/1756-8935-8-6

49. Bates D, Mächler M, Bolker B, Walker S. Fitting linear mixed-effects models using lme4. *J Stat Softw*. 2015;67(1):1-48. doi:10.18637/jss.v067.i01

50. Engdahl E, Alavian-Ghavanini A, Forsell Y, Lavebratt C, Rüegg J. Childhood adversity increases methylation in the *GRIN2B* gene. *J Psychiatr Res*. 2021;132:38-43. doi:10.1016/j.jpsychires.2020.09.022

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51. Okajima D, Kudo G, Yokota H. Antidepressantlike behavior in brain-specific angiogenesis inhibitor 2–deficient mice. *J Physiol Sci*. 2011;61(1):47-54. doi:10.1007/s12576-010-0120-0

52. Trivedi MS, Abreu MM, Sarria L, et al. Alterations in DNA methylation status associated with Gulf War illness. *DNA Cell Biol*. 2019;38(6):561-571. doi:10.1089/dna.2018.4469

53. Finkel Z, Esteban F, Rodriguez B, et al. *AAV6*-mediated Gsx1 expression in neural stem progenitor cells promotes neurogenesis and restores locomotor function after contusion spinal cord injury. *Neurotherapeutics*. 2024;21(4):e00362. doi:10.1016/j.neurot.2024.e00362

54. Melas PA, Wei Y, Wong CCY, et al. Genetic and epigenetic associations of *MAOA* and *NR3C1* with depression and childhood adversities. *Int J Neuropsychopharmacol*. 2013;16(7):1513-1528. doi:10.1017/S1461145713000102

55. Tyrka AR, Price LH, Marsit C, Walters OC, Carpenter LL. Childhood adversity and epigenetic modulation of the leukocyte glucocorticoid receptor: preliminary findings in healthy adults. *PLoS One*. 2012;7(1):e30148. doi:10.1371/journal.pone.0030148

56. Breton CV, Marsit CJ, Faustman E, et al. Small-magnitude effect sizes in epigenetic end points are important in children's environmental health studies: the Children's Environmental Health and Disease Prevention Research Center's Epigenetics Working Group. *Environ Health Perspect*. 2017;125(4):511-526. doi:10.1289/EHP595

57. Czamara D, Tissink E, Tuhkanen J, et al. Combined effects of genotype and childhood adversity shape variability of DNA methylation across age. *Transl Psychiatry*. 2021;11(1):88. doi:10.1038/s41398-020-01147-z

58. Lam D, Ancelin ML, Ritchie K, Freak-Poli R, Saffery R, Ryan J. Genotype-dependent associations between serotonin transporter gene (*SLC6A4*) DNA methylation and late-life depression. *BMC Psychiatry*. 2018;18(1):282. doi:10.1186/s12888-018-1850-4

59. Klengel T, Mehta D, Anacker C, et al. Allele-specific *FKBP5* DNA demethylation mediates gene-childhood trauma interactions. *Nat Neurosci*. 2013;16(1):33-41. doi:10.1038/nn.3275

60. Marini S, Davis KA, Soare TW, et al. Adversity exposure during sensitive periods predicts accelerated epigenetic aging in children. *Psychoneuroendocrinology*. 2020;113:104484. doi:10.1016/j.psyneuen.2019.104484

61. Tang R, Howe LD, Suderman M, Relton CL, Crawford AA, Houtepen LC. Adverse childhood experiences, DNA methylation age acceleration, and cortisol in UK children: a prospective population-based cohort study. *Clin Epigenetics*. 2020;12(1):55. doi:10.1186/s13148-020-00844-2

62. Lawn RB, Anderson EL, Suderman M, et al. Psychosocial adversity and socioeconomic position during childhood and epigenetic age: analysis of 2 prospective cohort studies. *Hum Mol Genet*. 2018;27(7):1301-1308. doi:10.1093/hmg/ddy036

63. Kim K, Yaffe K, Rehkopf DH, et al. Association of adverse childhood experiences with accelerated epigenetic aging in midlife. *JAMA Netw Open*. 2023;6(6):e2317987. doi:10.1001/jamanetworkopen.2023.17987

64. Meier M, Kantelhardt S, Gurri L, et al. Childhood trauma is linked to epigenetic age deceleration in young adults with previous youth residential care placements. *Eur J Psychotraumatol*. 2024;15(1):2379144. doi:10.1080/20008066.2024.2379144

65. Popham CM, McEwen FS, Karam E, et al. Predictors of psychological risk and resilience among Syrian refugee children. *J Child Psychol Psychiatry*. 2023;64(1):91-99. doi:10.1111/jcpp.13670

66. Moran JK, Jesuthasan J, Schalinski I, et al. Traumatic life events and association with depression, anxiety, and somatization symptoms in female refugees. *JAMA Netw Open*. 2023;6(7):e2324511. doi:10.1001/jamanetworkopen.2023.24511

67. Tinghög P, Malm A, Arwidson C, Sigvardsdotter E, Lundin A, Saboonchi F. Prevalence of mental ill health, traumas and postmigration stress among refugees from Syria resettled in Sweden after 2011: a population-based survey. *BMJ Open*. 2017;7(12):e018899. doi:10.1136/bmjopen-2017-018899

68. Danoff JS, Ramos EN, Hinton TD, et al. Father's care uniquely influences male neurodevelopment. *Proc Natl Acad Sci U S A*. 2023;120(31):e2308798120. doi:10.1073/pnas.2308798120

69. Brody GH, Yu T, Chen E, Beach SRH, Miller GE. Family-centered prevention ameliorates the longitudinal association between risky family processes and epigenetic aging. *J Child Psychol Psychiatry*. 2016;57(5):566-574. doi:10.1111/jcpp.12495

70. Sullivan ADW, Bozack AK, Cardenas A, et al. Parenting practices may buffer the impact of adversity on epigenetic age acceleration among young children with developmental delays. *Psychol Sci*. 2023;34(10):1173-1185. doi:10.1177/09567976231194221

71. Simpkin AJ, Howe LD, Tilling K, et al. The epigenetic clock and physical development during childhood and adolescence: longitudinal analysis from a UK birth cohort. *Int J Epidemiol*. 2017;46(2):549-558. doi:10.1093/ije/dyw307

72. Prebeg Z, Bralić I. Changes in menarcheal age in girls exposed to war conditions. *Am J Hum Biol*. 2000;12(4):503-508. doi:10.1002/1520-6300(200007/08)12:4<503::AID-AJHB10>3.0.CO;2-H

73. Li L, Denholm R, Power C. Child maltreatment and household dysfunction: associations with pubertal development in a British birth cohort. *Int J Epidemiol*. 2014;43(4):1163-1173. doi:10.1093/ije/dyu071

74. Stenson AF, Michopoulos V, Stevens JS, Powers A, Jovanovic T. Sex-specific associations between trauma exposure, pubertal timing, and anxiety in Black children. *Front Hum Neurosci*. 2021;15:636199. doi:10.3389/fnhum.2021.636199

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